

Accurate Humans in Virtual Worlds

Computer generated graphics are becoming ubiquitous in our modern world. To name a few examples, almost every film now utilizes CGI, video games have become one of the most popular forms of entertainment, and the medical industry heavily relies on 3D modeling software for diagnosis and treatment. These artificial worlds are cropping up everywhere, and they're populated by us, humans.

This presents the long-standing challenge of how to create an accurate computer model of the human body. While much of the focus historically has been reaching a level of photorealism that escapes the [uncanny valley](#), a less examined but equally important consideration is the representation of a diverse range of body types. In the current global context, ethical computing means that graphical modeling techniques of humans should aim to accurately represent all types of people in a fair and equal way globally.

Computer models of humans are brought to life with complex algorithms that simulate photorealistic skin and hair. It's important to recognize that within the origins of these algorithms lie some racial biases. Yale Computer science professor, Theodore Kim, has been one of the first professionals in the industry to [address a perceived historical bias](#) in human modeling techniques.

He explains how the dominant algorithm for modeling human skin replicates a physical phenomenon called "subsurface scattering." This can be seen in the subtle glow and translucency of white skin. But herein lies the issue. Black skin doesn't exhibit this effect. In fact it's typically characterized by specular highlights (shiny spots on a surface) which aren't realistically depicted by this model.

Similarly, he describes how the Marschner model for depicting hair works well for type 1 and type 2 hair. These types are straight and wavy, those typically held by white people. It doesn't work well for type 4 or afro-textured hair which is coiled and kinky.

It wasn't just the inaccurate algorithms that Kim took issue with but the very language of his field. He was working in the graphics industry in 2002 when Peter Jackson's [The Two Towers](#) was released. The innovative use of Stanford's subsurface scattering model on the famous representation of Gollum quickly made this phenomenon synonymous with skin in the industry. This notion is obviously problematic as it really only represents *white* skin. He's also brought attention to the fact that white models are almost exclusively used in research papers and industry documentation when talking about modeling skin or hair.

Kim does take a rather extreme position, explicitly calling the history of computer animation "racist" in his [opinion piece for Scientific American](#). His rhetoric very clearly stems from that of the Black Lives Matter era of 2021, focusing on historical bias in American culture and the idea that systemic racism is inseparable from all aspects of our lives, including math and physics.

Some other professionals were less than pleased with this notion, but social issues aside, the discrepancies that Kim addresses are inaccuracies that should be rectified.

Luckily, the graphics industry has recently moved towards using a more general physically-based rendering model for all materials including skin. As described by manufacturer [NVIDIA](#), "Simple empirical specular calculations, such as the familiar Blinn-Phong model ... do not accurately approximate the specular reflectance of skin. Physically based specular models provide more accurate-looking results, leading to more realistic images."

Hair modeling is also rapidly evolving as older models were primitive and visually unappealing. [This paper](#) is an example of a newer model that was tested using curlier types of hair. These minor improvements in the equality of modeling techniques likely won't dismantle decades of implicit bias, but at the very least they're a step in the right direction.

It's easy to see where these inaccuracies originated from. The original developers of these models were likely white, and created accurate models to depict their own body. They may not have had much personal experience with black hair or skin, and were piggybacking off a century of a white-dominated film industry standards. But that was many years ago, and in the 21st century there's no reason a computer model can't accurately portray any type of human being.

The industry as a whole should aim to adopt an inclusive model for human skin and hair that will fairly include all global peoples. This may take the form of a one size fits all model as is used in physically-based rendering, or may require the development of separate algorithms for dark and light skin. Whichever approach promotes the most algorithmic fairness should be chosen.

The [Code of Ethics](#) released by the ACM (Association for Computing Machinery) is the set of guiding principles for a developer in the modern age. Principle 1.4 states, "Be fair and take action not to discriminate," followed by, "Computing professionals should foster fair participation of all people, including those of underrepresented groups." By this principle, graphics developers have an obligation to create fair and accurate models of all types of human bodies.

Because virtual worlds are becoming so popular in everyday life, they largely affect our view of the world. Misrepresentation of specific groups doesn't just harm them, but also the global perception of those communities. If this medium is biased or inaccurate when representing people of color, it may rub off on a larger societal scale by skewing the cultural perception of them. This breeds the same implicit bias that created the problem, creating a self-affirming loop that can only be broken by a conscious effort and decision to correct past errors.

We are not perfect, and our past is riddled with inequality, but we must pursue an ideal of ethical computing. The segregation of math and computing from social issues may have inadvertently led us to this point, but it provides us an opportunity for change. Professor Kim summarizes this well, "The idea that math and science provide an objective way to understand the world and ourselves is not an ironclad guarantee, but rather an ideal to strive for."

Works Cited

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